

Informatics and computer science, Winter 2025
“ Comprehensive Human Resource Management System (HRMS)”
Milestone 3
Deadline: 14/12/2024

Requirements:

In this milestone, you are required to implement a website that connects to your database to perform the different functionalities described in the project.

Deliverables:

You are required to submit the following:

- **.zip** file, named after your team number, having the following documents:
 - Source code of the submitted website: All the source code files needed to run the project.
 - SQL Script of the implemented Database: SQL file containing the script of the data of the database that you have implemented on SQL Server (submit the script (Milestone 2) if you haven't modified anything).
 - A PDF file containing :
 - * Team members' names.
 - * Team members' tutorial groups.
 - * Team members' application numbers.
 - * Team members' chosen component.

Submission guidelines:

- **No late submissions** will be accepted under any circumstances.
- In case no one submitted from a certain team the whole team will be assigned a zero in milestone 3.

Read the following carefully:

- In this document, we divided the website into five components and a general component. All components should be **integrated** together. If your team is less than five members then two of the components has to be the profile and courses components. Each team member should be assigned to one component to implement.
** The General Component is a MANDATORY one and all the team members will be graded on general component.
- Each team is required to choose the component that he/she will be working on and add it to the PDF file as mentioned in the Deliverables.
- No team members can be assigned to the same component.
- Each team member will be evaluated and graded according to the component he/she was assigned to + general component which means that each member is completely responsible for his/her component.

- If your team is less than 5 members you don't have to cover all components. For example, if you are only 3 members, then you have to choose the first 3 components, where each member will be assigned to a component.
- The General Component will be implemented by along with component 4 as a single person component.
- Teams with good User interface will get a bonus.

Web Application Components

Note that:

- a) You have to show whether any action on the website was successful or not. In case of any unsuccessful action, you need to notify the user with the reason and the steps needed to correct the action.
- b) If a stored procedure/function from Milestone 2 was incorrect or missing any input/output, you have to correct it before using it in this milestone.
- c) The items of each component should be integrated with each other. Furthermore, all components should be integrated together. In other words, we should be able to navigate through your website from one functionality to another by interacting with the web pages of the website and NOT by changing the URL.

General Component:

- a) System Admins, HR Admins and Line Managers can create their personal accounts by providing required details such as email, password, role, and profile information.
- b) System admins can create accounts to new employees by providing required details such as email, password, role, and profile information.
- c) All employees should be able to log in to their accounts using their credentials.
- d) HR Admins can edit any part of employees profile.

Bonus

1. Users can upload a profile picture to their account.

Component 1- Employee Profiles Contracts Management:

This component includes all pages required to view, edit, and manage employee data, roles, hierarchy, and contract records.

Employee Profile Management

- a) Admins and Managers can view full employee profiles.
- b) Employees can update personal details and emergency contacts.
- c) System Admins can assign system roles (Employee, HR Admin, Manager) and view all employees in all departments.
- d) Managers can view their team details.
- e) HR Admin can manage the completeness of Employees profile.

Contract Management

- a) HR Admins can create employment contracts for employees.

- b) HR Admins can renew or update expiring contracts.
- c) The system should list active and soon to expire contracts.
- d) Contract updates must trigger notifications.

Component 2- Attendance Shift Management:

This component includes all pages and logic related to attendance recording, shift creation, and exception rules.

Shift Management

- a) System Admins can create shift types.
- b) HR Admins can configure split shifts and rotational shifts.
- c) System Admin and Manager can assign shifts to employees or entire departments.
- d) System Admin can assign and update normal, rotational and custom or special shifts to individual employees.

Attendance Tracking

- a) Employees can view and record daily attendance through the system.
- b) Employees can submit attendance correction request.
- c) System Admin can sync leave with attendance system.
- d) Offline attendance logs must sync when the user reconnects.
- e) The system must apply grace periods, short-time penalties, and time rules.
- f) Managers can view their teams attendance summary.

Component 3- Leave Management System:

This component handles all HR leave-related operations.

Employee Leave Features

- a) Employees can submit leave requests with type, dates, and attachments.
- b) Employees can view their leave history and remaining leave balance.

Manager Leave Features

- a) Managers can view pending leave requests from their team.
- b) Managers can approve or reject leave requests.
- c) Managers can flag irregular leave patterns.

HR Admin Leave Features

- a) HR Admins can configure leave types, policies, and eligibility rules.
- b) HR Admins can assign or adjust leave entitlements.
- c) HR Admins can override a leave decision if necessary.
- d) HR Admins can manage special leave types (e.g., maternity leave).
- e) Leave approvals must sync automatically with attendance records.

Component 4- Mission & Task Management

This component includes system alerts, organization visualization, and analytical insights.

- a) Employees can view assigned missions.
- b) Managers can approve or reject mission requests.
- c) HR Admins can assign mission to Employees.

Component 5- Notifications, Analytics & Hierarchy Dashboard

This component includes system alerts, organization visualization, and analytical insights.

Notifications

- a) Users receive notifications for contract expirations, leave approvals, shift reassignments, and mission updates.
- b) Line Managers can send customized notifications to team members.
- c) All Employee should be able to view their notifications.

Analytics Reporting

- a) HR Admins can generate department-wise employee statistics.
- b) HR Admin should be able to search and generate compliance or diversity reports.
- c) All employees should be able to login into their account to access their profiles.

Hierarchy Dashboard

- a) System Admins can view the entire organizational hierarchy.
- b) System Admins should be able to reassign an employee to a new department or manager within the hierarchy.
- c) Users can navigate through departments, managers, and teams visually.